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TITLE PLACEHOLDER

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Abstract

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ABSTRACT PLACEHOLDER

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Introduction

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Suppose that you attend a friend’s wedding. That evening you describe it to your partner, re-
counting how the ceremony ran long, how the best man’s toast was unexpectedly moving, and
how a small child wandered onto the dance floor midway through the first song. A week later a
coworker asks how the wedding was, and you tell the story again. What is that second telling a
telling of? One appealing answer is that both tellings draw on the same underlying memory of
the wedding, and that whatever differences arise between them reflect the ordinary imprecision of
putting an experience into words. A second possibility, which motivates the work we report here,
is that your first telling has itself become part of what you are remembering.

14

What happens to an experience between living it and describing it later? Since our brains
do not maintain an exhaustive account of everything we encounter, the very act of perceiving an
experience already necessarily entails compressing and distorting it. We attend closely to some

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17 aspects of a scene while ignoring others, and we tend to interpret ambiguous moments in light
18 of what has happened previously and we expect will happen next. But once an experience has
19 ended, neither the experience itself nor our moment-to-moment perception of it remains available,
20 to us or to anyone else. What remains is instead the product of a further distillation, in which
21 our perception of the experience is reduced to something our memory systems can store and
22 later reconstruct. Which aspects of an experience survive this second distillation may depend on
23 what we attended to, what we anticipated would matter later, what we found interesting, and the
24 thoughts, motivations, and biases we brought with us. Because remembering entails reconstructing
25 an experience rather than replaying it, the resulting account often preserves the general “shape”
26 of what happened while discarding many of its particulars (Bartlett, 1932; Schacter, 2012; Schacter
27 et al., 2011). The units over which these distillations operate appear to be events, in that people
28 segment continuous experience into discrete episodes at moments of change, and those boundaries
29 shape what is remembered afterwards (Baldassano et al., 2017; Radvansky and Zacks, 2017). In
30 prior work we characterized this transformation geometrically by modeling an experience and a
31 person’s recounting of it as trajectories through a semantic space, and asking which properties of
32 the experience’s trajectory are preserved in the recounting (Heusser et al., 2021).

33 Yet memories do not remain fixed once they have been encoded, pointing to further transfor-
34 mations that a single recounting cannot reveal. Information may be lost over time; experiences we
35 have after an event may color how we later remember it; similar experiences may interfere with
36 one another; and we may simply re-evaluate what happened in light of what we have learned
37 since. The act of retrieving a memory itself appears to be one of the forces that changes it, in that
38 a memory can become labile when it is reactivated and must be stabilized again afterwards (Hup-
39 bach et al., 2007, 2009; Nader et al., 2000; Sederberg et al., 2011). Retrieval also changes what will
40 be accessible later—often substantially—relative to memories that go unretrieved (Karpicke and
41 Roediger, 2007; Roediger and Karpicke, 2006). Further, the particular way in which we recount
42 an experience appears to matter as well, since people who are asked to describe an event for one
43 purpose later remember it in ways that are biased toward that description (Tversky and Marsh,
44 2000). Memories also become more semantic and less perceptually detailed, both with the passage

45 of time and with repeated remembering (Lifanov et al., 2021). Repeated retrieval of naturalistic
46 events over the course of a week has been found to stabilize each person’s recall (for example,
47 which details they include) without causing different people’s recalls to converge (Melega et al.,
48 2025). Taken together, these observations suggest that each time we remember an experience, the
49 substance of the underlying memory we are retrieving may differ from what we retrieved before.

50 How might we compare two recountings of the same experience when the experience and
51 both recountings take the form of unconstrained natural language? Recent work has begun to use
52 language models to quantify the content of what individuals say when they recount what they
53 remember (Fenerci et al., 2025; Fitzpatrick et al., 2026; Heusser et al., 2021; Manning et al., 2022).
54 The approach developed in our prior work embeds the description of an experience, and each
55 individual’s recounting of it, into a common high-dimensional space whose coordinates reflect
56 the concepts under consideration, and through which the experience and recountings each unfold
57 over time as a geometric trajectory (Fitzpatrick et al., 2026; Heusser et al., 2021; Vera et al., 2025).
58 We then segment the experience and each recounting into events and identify which event from
59 the experience each recall event describes, yielding sequences of events that may be compared
60 directly across people and across delays (Baldassano et al., 2017; Chen et al., 2017; Heusser et al.,
61 2021).

62 Here we apply our approach to data from an experiment in which participants viewed an
63 episode of a television show, recounted its events immediately afterward, and then recounted
64 them again following a week-long delay. Because we collected two recountings of the same
65 experience from each participant, we are able to ask what participants’ delayed recountings most
66 closely resemble. If retrieval recovered the original experience in an uncorrupted form, then both
67 recountings should be about equally similar to the experience itself. We find instead that the
68 structure of a participant’s delayed recounting is more similar to that of their own immediate
69 recounting than to that of the original episode. It is also more similar to their own immediate
70 recounting than to other participants’ immediate recountings, to the degree that an individual’s
71 immediate recall can be identified from their delayed recall on the basis of this structure alone.
72 What governs retrieval after a delay therefore appears to be a person-specific distillation, rather

73 than the episode as it was first experienced.

74 Quantifying the semantic geometry of individuals' recalls at different points in time offers the
75 opportunity to characterize the transformations undergone by our memories beyond an initial
76 recounting. For example, if individuals' recountings tend to retain their idiosyncratic structures
77 after a delay, how *do* the structures of our memories tend to change with time? There is a particular
78 qualitative change captured in the ways we tend to describe how memories change

79 There is a particular qualitative change our memories undergo as we naturally forget aspects
80 of experience, implied by common descriptors such as "fading", "blurring", becoming "fuzzy", or
81 "decaying" (Thorndike, 1913). But the nature of this change can be difficult to characterize more
82 concretely. Here we show that this transformation is well described as geometrically simplifying
83 the path through semantic space a participant follows as they recount an experience, in that
84 applying existing for removing detail from a geometric curve

85 Results

86 We

87 *[OUTLINE — placeholder structure, not final prose. Bracketed italic text marks content still*
88 *to be written. No statistic below has been computed or verified; every "report [STAT]" is a hole to*
89 *be filled from the cited notebook's output.]*

90 *[FIGURE BUDGET: 8 main-text figures maximum. The plan below uses 6, leaving 2 in re-*
91 *serve (one likely wanted for the RDP analysis if it needs a second panel set, one for contin-*
92 *gency). Figure 1 (paradigm.pdf) is the only finished main-text figure; the other PDFs already in*
93 *paper/figures/source/ are analysis outputs that may feed panels rather than figures in their*
94 *own right. Nothing else is made yet.]*

95 From episode and recalls to comparable event sequences

96 *[Walk the reader through the pipeline, briefly and conceptually (details live in the Methods), up to the point*
97 *where we have three things in hand: (a) the sequence of events comprising the episode, (b) each participant's*

98 sequences of immediate and delayed recall events, and (c) the mapping between each recall event and the
99 episode event it describes. This is the setup that everything below depends on — the point to land is that the
100 episode and both recalls now live in a common space and are segmented into comparable units.]

101 [Figure 1 — **paradigm**. `paradigm.pdf`, already in `paper/figures/source/`. Session-1 row: in-
102 structions, episode (24 min), immediate recall (10–60 min), prediction (5 min); one-week gap; session-2 row:
103 instructions, delayed recall (10–60 min), session-2 episode (20/22 min), recall of that episode (10–60 min).
104 Planned revision: reduce the opacity of the blocks for the phases we don’t analyze — the session-1 prediction
105 block, and the session-2 episode-viewing and subsequent-recall blocks — so the reader’s eye lands on the
106 immediate/delayed recall pair that the paper is actually about.]

107 [Figure 2 — **methods overview**. To be made. Annotations and recall text → trajectories → event-
108 segmentation procedure → event sequences. Loosely modeled on Fig. 2 of Heusser et al. (2021), extended
109 with the (new) event-segmentation step. Layout still open. Candidate source material already generated:
110 `episode-corrmat.pdf` and `k-optimization-episode.pdf` (from `event-segmentation.ipynb`), and
111 the per-recall similarity heatmaps and match matrices at `lep1-and-delayed-recall-similarity-heatmaps.pdf`
112 `/-matchmats.pdf`.]

113 **Participants recall the episode’s events in a similar order before and after the** 114 **delay**

115 [Motivation: we wanted to know whether participants remembered the episode similarly before and after the
116 delay. The simplest way to ask this is whether they recalled the same events in the same order.]

117 [Result: recall-order similarity. Computed in `self-similarity-analyses.ipynb`, section “Is de-
118 layed recall order more similar to own immediate recall order than to others’ immediate recall orders?”. The
119 measure compares pairwise precedence relations over the episode events shared between two recalls, then
120 percentile-ranks each participant’s delayed-vs-own-immediate similarity against their delayed-vs-others’-
121 immediate similarities, tested against chance (0.5) with a Wilcoxon signed-rank test. Report [STAT].]

122 [Figure 3 — **recall-order similarity**. To be made; multi-panel.]

123 [Panel A: illustration of the method. For one example participant, show the matrix of pairwise recall-

124 *order-precedence values over pairs of episode events — probably just the lower triangle — for their delayed*
125 *recall beside their own immediate recall. Grey out (or reduce the opacity of) cells for event pairs not recalled*
126 *in both sessions, since the comparison is restricted to pairs present in both. Then show the same thing again*
127 *for that participant's delayed recall against a different participant's immediate recall, which is the contrast*
128 *the analysis actually tests. This maps directly onto `pairwise_precedence()` and the `shared_events`*
129 *argument to `order_similarity()` in the notebook, so the panel and the code should be checked against*
130 *each other.]*

131 *[Panels B–D: de-sloped versions of panels B–D of the reference mock-up. Include the analogues of that*
132 *mock-up's B and D so that the same panel types can reappear in the recall-structure figure (Figure 5), letting*
133 *the reader compare order and structure on identical terms — this parallelism is the reason for the pairing*
134 *and should be preserved in the final layout.]*

135 *[Note on ordering: this is the least novel analysis of the three and the one least dependent on our*
136 *framework — order could be measured without any embedding model. It goes first precisely because it*
137 *establishes the basic effect on the simplest possible terms, setting up the contrast with what follows.]*

138 **The semantic structure of participants' recalls is preserved across the delay**

139 *[Motivation: order is only the coarsest description of a recall. Because our approach embeds recalls in a*
140 *continuous semantic space, we can ask a stronger question — whether the structure of how a participant*
141 *moved through the episode's content is preserved, not merely the sequence in which they hit the events.]*

142 *[Figure 4 (optional) — 2D trajectories. Decision deferred, and it depends on how this section's*
143 *narrative settles. Argument for including it: it introduces the idea of geometric comparison and establishes*
144 *the 2D paths before they get used as examples in Figure 5, so the reader isn't meeting the representation*
145 *and the result at the same time. Argument against: it spends one of 8 figure slots on setup. Middle*
146 *option — a compact version showing only the episode path plus immediate and delayed paths for a handful*
147 *of participants, with the full version relegated to the supplement. Source: `2d-trajectories.ipynb`*
148 *(`atlep1-paths2d.pdf` and `delayed-paths2d.pdf`, `imm-del-2d.pdf`); note those `savefig` calls are*
149 *currently commented out in the notebook.]*

150 *[Result: trajectory/event-structure similarity. Computed in `self-similarity-analyses.ipynb`,*

151 sections “Is delayed recall more similar to own immediate recall than to episode?” and “Is delayed recall
152 more similar to own immediate recall than to others’ immediate recalls?”. Both use DTW warp cost (average
153 cosine distance between matched events) over mean-centered event sequences; the second percentile-ranks
154 self against others and tests against chance. Report [STAT] for each.]

155 [Framing: this result appears strong enough to present as a “fingerprint” — i.e., that one can identify
156 which immediate recall belongs to a given participant from the structure of their delayed recall alone. Frame
157 it that way in both the text and the figure, but confirm the effect size supports the claim before committing
158 to that language.]

159 [Figure 5 — **recall-structure similarity**. To be made; the largest figure in the paper, covering both
160 analysis (A) “delayed is more similar to immediate than to the episode” and analysis (B) “delayed is more
161 similar to own immediate than to others’ immediates”. Layout still open.]

162 [Example trajectory panels (3, from the current draft): the left panel is common to both analyses, so
163 it should sit centrally in the final arrangement; the right panel is the contrast for analysis (A) and the
164 middle panel is the contrast for analysis (B). Planned refinements: thicker lines; faint connectors between
165 DTW-matched events within each panel (already present in the draft); possibly PCHIP splines rather than
166 straight segments between points, to make the paths read as curves. Under consideration — a fourth panel
167 showing pairwise mean point-by-point distances between trajectories, which would require upsampling each
168 trajectory to a common number of points and comparing them in order (or substituting DTW cost, if that
169 turns out to read better).]

170 [Analysis (A) panel: per-participant connected dots showing each participant’s delayed-to-immediate
171 versus delayed-to-episode alignment cost. Probably without the KDEs from the draft version.]

172 [Analysis (B) panels (3): de-slopified versions of the current draft. Two of them are the direct analogues
173 of Figure 3’s panels B and D and show stronger effects — keeping the panel types visually identical across
174 the two figures is what makes “structure beats order” legible without the reader doing arithmetic. The third
175 gets at how person-specific the similarity is, and is the panel that carries the “fingerprint” claim.]

176 [Candidate supporting material: `atlep1-recall-corrstats.pdf` and `delayed-recall-corrstats.pdf`
177 (from `event-segmentation.ipynb`).]

178 **Delayed recalls are simplified versions of participants' immediate recalls**

179 *[Motivation: having established that a participant's delayed recall structure resembles their own immediate*
180 *recall structure, the natural next question is how it changes over the delay.]*

181 *[Result: trajectory simplification. Computed in `rdp-analyses.ipynb`. Each participant's immediate*
182 *recall trajectory is simplified with the Ramer-Douglas-Peucker algorithm over a range of tolerances (ϵ), and*
183 *the ϵ minimizing DTW warp cost to their delayed recall trajectory is selected. Sub-results to report: (i) the*
184 *relationship between how many events a participant dropped across the delay and how many RDP removes*
185 *at the best-fitting ϵ (Kendall's τ); (ii) the same against ϵ itself; (iii) whether RDP predicts which events*
186 *were forgotten, benchmarked against a hypergeometric null and compared to predictions derived from other*
187 *participants' trajectories. Report [STAT] for each.]*

188 *[Figure 6 — RDP simplification. To be made, and how best to illustrate the simplification is genuinely*
189 *unresolved — flagged as an open design question rather than a layout still to be drawn. The regression plots*
190 *for sub-results (i) and (ii) exist in `rdp-analyses.ipynb` but have no `savefig` calls, so nothing is exported*
191 *to `paper/figures/source` yet. One thing worth trying: since Figure 5 will already have taught the reader*
192 *to read 2D trajectories, showing one participant's immediate path beside its RDP-simplified version and*
193 *their actual delayed path would make “the delayed recall looks like a simplified version of the immediate one”*
194 *visible directly, with the quantitative panels as support. If the analysis needs a second panel set, one of the 2*
195 *reserve figure slots can absorb it.]*

196 *[@paxton: does the simplification story want a fourth subsection on what the dropped events have in*
197 *common (e.g. are they lower-salience, more peripheral to the narrative), or is that a Discussion point?]*

198 **Discussion**

199 **Methods**

200 Our experimental protocol was approved by the Committee for the Protection of Human Subjects
201 at Dartmouth College. We obtained written informed consent from all participants prior to the
202 start of both experiment sessions.

203 **Participants**

204 We enrolled a total of 70 Dartmouth undergraduate students in our study. Participants received
205 optional course credit for enrolling and were pre-screened to ensure they had not previously seen
206 any episode of either TV show viewed during the experiment. No statistical method was used
207 to predetermine sample size. Prior to data processing and analysis, we excluded a total of 17
208 participants from our dataset: 4 encountered technical issues with the experiment, 2 self-reported
209 difficulty with the task due to illness, 4 spoke for less than 10 minutes during the immediate recall
210 phase, and 7 did not return for the second experiment session. The data we analyzed comprised
211 the remaining 53 participants.

212 At the start of the experiment, we asked each participant to complete a demographic survey
213 that included questions about their age, sex, race, and ethnicity. Participants' ages ranged from 18
214 to 22 years (mean: 18.91 years; standard deviation: 1.01 years). A total of 41 participants reported
215 their sex as female and 12 participants reported their sex as male. Participants reported their races
216 as White (34 participants), Asian (16 participants), Black or African American (3 participants),
217 American Indian or Alaska Native (1 participant), and other (1 participant). (Note that some
218 participants selected multiple racial categories.) A total of 52 participants reported their ethnicity
219 as "Not Hispanic or Latino" and 1 participant reported their ethnicity as "Hispanic or Latino."

220 We also asked participants their native spoken language; whether they had any hearing, speech,
221 or color vision impairments; and about any current medications or recent injuries that could impact
222 attention or memory. All participants reported that their native spoken language was English, that
223 they had no hearing or speech impairments, and that they had normal color vision. A total of 52
224 participants reported no attention- or memory-relevant medications or injuries. One participant
225 reported sustaining a concussion more than 3 years prior to the experiment and was not excluded
226 from our dataset.

227 Additionally, before each of the two experiment sessions, we asked participants about their
228 sleep, coffee consumption, and level of alertness. Participants reported having slept between 5
229 and 11 hours the night prior to session 1 (mean: 7.37 hours; standard deviation: 1.45 hours) and

between 5 and 11 hours the night prior to session 2 (mean: 7.39 hours; standard deviation: 1.23 hours). They reported having consumed between 0 and 3 cups of coffee on the day of session 1 (mean: 0.43 cups; standard deviation: 0.72 cups) and between 0 and 3 cups of coffee on the day of session 2 (mean: 0.48 cups; standard deviation: 0.68 cups). Participants reported their current levels of alertness on a 5-point Likert scale. At the start of session 1, 1 participant reported feeling "Very sluggish," 14 reported feeling "A little sluggish," 18 reported feeling "Neutral," 15 reported feeling "A little alert," and 5 reported feeling "Very alert." At the start of session 2, 3 participants reported feeling "Very sluggish," 9 reported feeling "A little sluggish," 23 reported feeling "Neutral," 17 reported feeling "A little alert," and 3 reported feeling "Very alert."

@jeremy: do we want to report class years and majors? And/or do we want to report post-experiment survey responses (how engaging did you find the episode, how easy/difficult was it to follow, how well do you think you recalled it, how well do you think you learned characters' names)?

Experiment

The experiment took place over two sessions lasting approximately one hour each, separated by 6–8 days (mean: 6.76 days; standard deviation: 0.73 days). Prior to the start of the experiment, participants were assigned to one of two groups (*A* or *B*), alternating based on enrollment order. During session 1, all participants first viewed the pilot episode of the FX series *Atlanta* ("The Big Bang"; duration: 24 min 15 s). Immediately after viewing, participants were given up to one hour to recount what happened in the episode to the best of their abilities. Following Chen et al. (2017), participants were instructed to describe the episode in as much detail as possible and to try to speak for a minimum of 10 minutes. They were told to try to recount the episode's events in their original order, but that they were free to go back and describe earlier events should they realize they had missed something. After the immediate recall phase, participants were then given up to 5 minutes to predict what they thought would happen in the show's next episode.

Upon returning for session 2, all participants were again given up to one hour to recount the *Atlanta* episode they had viewed the week prior. Following the delayed recall phase, participants in Group *A* then viewed the second episode of *Atlanta* ("Streets on Lock"; duration: 21 min

57 s) while participants in Group *B* instead viewed the pilot episode of the FOX series *Arrested Development* (“Pilot”; duration: 20 min 37 s). Both groups were then given up to one hour to recount the just-watched episode in the same manner and given the same instructions as in session 1. We implemented the experiment using the psiTurk platform (Gureckis et al., 2016) and jsPsych library (de Leeuw, 2015), with a custom jsPsych plugin we developed in prior work for recording verbal recall audio (Ziman et al., 2018). Here we analyze participants’ immediate and delayed recalls of the *Atlanta* episode viewed in session 1.

Stimulus annotation and recall transcription

To capture the evolving semantic content of the *Atlanta* episode, we created detailed text annotations of the on-screen content during each individual camera shot throughout the episode (mean annotation duration: 3.05 s; standard deviation: 1.80 s). Specifically, we used ANVIL (Kipp, 2001) to record, for each camera shot, a set of semantically relevant features. These included a brief description of physical occurrences and characters’ explicit actions, a brief description of characters’ internal states (e.g., emotions, motivations) and other implicit content, the names of all characters appearing on screen, a transcription of any dialogue, the name(s) of the character(s) speaking, the shot’s setting, whether the shot took place indoors or outdoors, and whether or not there was music playing during the shot. This feature set was adapted from a similar annotation corpus created by Chen et al. (2017), which we used in developing our trajectory-analysis framework in prior work (Heusser et al., 2021).

We then transcribed each participant’s immediate and delayed verbal recall of the *Atlanta* episode. Our transcription process consisted of three steps: an initial manual transcription, a two-pass prose-formatting conversion with a large language model, and a final manual cleanup and correction step. First, we manually transcribed each recall audio file collected during the experiment. These initial transcripts reflected participants’ spoken recalls verbatim, without any punctuation or additional formatting, and including disfluencies, stuttered repetitions, and other artifacts common to unrehearsed speech. However, since transformer models (such as we used to obtain text embeddings of participants’ recalls) are trained almost exclusively on written text,

284 structural conventions like sentence breaks, punctuation, and capitalization provide an important
285 signal these models leverage to contextual other tokens (e.g., words) and construct a representation
286 of the text’s overall meaning. Meanwhile, speech-native phenomena such as stuttered repetitions
287 and filler words (“like”, “um”, etc.) that are comparatively rare in written text contribute out-of-
288 distribution noise that can be expected to degrade embedding fidelity, and prior work has shown
289 that automated removal of these phenomena improves performance on downstream tasks that
290 assume clean, written-style input (Chen et al., 2022; Wang et al., 2010). We therefore used a large
291 language model (Qwen3.5 122b; Qwen Team, 2026) to convert participants’ recall transcripts to
292 a structure and style more typical of written text. We used the `langchain_dartmouth` Python
293 package (Stone et al., 2025) to implement this conversion as two sequential passes over each
294 transcript. In the first pass, we prompted the model to split the transcript into sentences at
295 natural breaking points, add punctuation including quotes around reported speech, remove clear
296 non-word disfluencies (“um,” “uh,” “hmm,” etc.), collapse stuttered verbatim repetitions, and
297 mark self-corrections and restarts with em-dashes. In the second pass, we prompted the model
298 to remove all instances of “like” used as a filler word. We removed filler “like” instances in a
299 separate pass because of their high frequency of occurrence in the transcripts and the additional
300 effort to required to distinguish them from meaningful uses of “like.” The full texts of our prompts
301 for both passes can be found in *Supplementary methods*. Finally, we manually reviewed all LLM-
302 formatted transcripts to identify and correct any errors, ensure sentence breaks were appropriate
303 and consistent across participants, confirm that no content was improperly added to or removed
304 from the transcripts, and resolve any ambiguities based on the prosody of participants’ original
305 recall audio.

306 **Analysis**

307 **Statistics**

308 Wilcoxon tests were one-sided because our alternative hypothesis was directional (delayed matches
309 own immediate *better than* the episode, others’ immediates, etc.).

310 Constructing episode and recall trajectories

311 We adapted a method we developed in prior work (Heusser et al., 2021) to transform the *Atlanta*
312 episode and participant’s recalls of it into trajectories through a common representation space
313 using a text embedding model. Briefly, text embedding models take in natural language text
314 of an arbitrary length and output a fixed-length vector representation that captures the text’s
315 semantic meaning. Intuitively, an output vector of length n can be viewed as a coordinate in an
316 n -dimensional text embedding space. While specific model architectures and training paradigms
317 vary, modern text embedding models are broadly trained to output similar coordinate vectors (in
318 terms of cosine similarity, dot product, inverse Euclidean distance, or other geometric measures) for
319 semantically similar input texts, even when those texts share few or no specific words. We obtained
320 text embeddings for the episode and recalls using EmbeddingGemma (Vera et al., 2025), a pre-
321 trained 308-million-parameter encoder-only transformer model that outputs coordinate vectors in
322 a 768-dimensional embedding space.

323 To construct the trajectory for the *Atlanta* episode, we first converted our tabular annotations
324 of the content present during each camera shot to a plain-text format. Separately for each shot,
325 we concatenated the annotation text describing explicit on-screen occurrences and characters’
326 implicit internal states. We then appended “Speech: ” followed by the shot’s dialogue, “Character
327 speaking: ” followed by the name of the speaker, and “Setting: ” followed by the shot’s setting. If
328 any annotated fields were empty for a given shot (e.g., if no character was speaking at that time), we
329 omitted them and their prefixes from the concatenated text. Next, we used EmbeddingGemma to
330 transform the text snippet for each camera shot into a text embedding vector, resulting in a number-
331 of-shots (478) by number-of-features (768) matrix. Since the camera shots varied in duration, we
332 then resampled this series of embedding vectors to a consistent temporal resolution. We assigned
333 each vector to the timepoint midway between the start and end of its corresponding camera shot
334 (in seconds), and used linear interpolation to estimate a vector for each second of the episode. This
335 yielded a number-of-seconds (1,455) by number-of-features (768) matrix—i.e., a 768-dimensional
336 *trajectory*.

337 We then created similar trajectories for each participant’s immediate and delayed recall of the
338 episode. We began by splitting each recall transcript into a list of sentences, with the constraint
339 that splits could not occur within reported speech (i.e., if a participant recounted a multi-sentence
340 quote by a character in the episode, we treated the full quote as a single sentence-unit). We
341 then parsed these sentences into overlapping sliding windows that spanned up to 5 consecutive
342 sentences each. These windows grew to and shrank from their full size at the beginning and end of
343 a participant’s recall sequence—in other words, the first window contained only the first sentence
344 of the participant’s recall, the second contained the first two sentences, and so on. This ensured
345 that all sentences from a participant’s recall transcript appeared an equal number of times (5) in
346 their sequence of sliding windows. Finally, we used EmbeddingGemma to transform each sliding
347 window into a text embedding vector. Repeated for all recall transcripts, this yielded a separate
348 trajectory characterizing each participant’s recall from each of the two experiment sessions. The
349 number of samples in participants’ recall trajectories (i.e., the number of sliding windows parsed
350 from their recall transcripts) ranged from 61 to 308 (mean: 136.68; standard deviation: 49.70).

351 **Relating episode and recall embedding vectors**

352 In prior work using our trajectory-analysis framework (e.g., Fitzpatrick et al., 2026; Heusser et al.,
353 2021; Manning, 2021; Manning et al., 2022; Racciah et al., 2024), embeddings of the stimulus and
354 participants’ responses were constructed using a topic model (Latent Dirichlet Allocation; Blei
355 et al., 2003) trained solely on the stimulus content. Here we elected to instead use a pre-trained
356 transformer-based text embedding model due to the richer and more nuanced textual information
357 these models are able to capture. (For example, these context-sensitive models can disambiguate
358 the texts “Earn looked at Alfred” and “Alfred looked at Earn” in their representations, whereas
359 bag-of-words models like LDA would represent the two identically.) However, these richer rep-
360 resentations also introduce certain geometric nuisances we needed to account for in order to
361 effectively relate the contents of participants’ recalls to that of the episode they viewed. First,
362 because modern transformer models are pre-trained on large and diverse text corpora, the repre-
363 sentational capacities of their embedding spaces span an enormous breadth of semantic content.

364 Consequently, in preliminary analyses we found that the embeddings of our annotations and par-
365 ticipants’ recalls (which were all “about” the events, themes, and characters of a single *Atlanta*
366 episode) were strongly anisotropic—i.e., they all occupied the same relatively narrow angular
367 region of the model’s overall embedding space. Additionally, because transformer models are sen-
368 sitive to the syntactic structure of text, their embedding representations carry latent information
369 about the text’s linguistic register in addition to its semantic content. Accordingly, we found there
370 was a consistent angular offset between embeddings of the episode annotations and those of par-
371 ticipants’ recalls, reflecting the fact that the annotations comprise grammatically well-structured
372 written text while the recalls comprise transcripts of casual unrehearsed speech.

373 We corrected for these representational artifacts by mean-centering the episode and recall trajec-
374 tories before performing analyses that involved measuring similarities between embedding vectors
375 within or across trajectories. Conceptually, mean-centering a given set of embedding vectors (i.e.,
376 subtracting the across-vector average from each individual vector) serves to remove representa-
377 tions that are common across all vectors under consideration, and instead highlight representations
378 that *vary* between them. For example, consider the sequence of embedding vectors comprising
379 the trajectory of the *Atlanta* episode. Common across vectors within this sequence will be latent
380 signals driven by how we chose to format the annotation text passed to the embedding model, the
381 annotator’s particular idiolect (i.e., idiosyncratic vocabulary and writing style), and any high-level
382 overarching themes of the episode. *Not* common across these vectors are the finer-grained semantic
383 representations that distinguish different events, scenes, and actions throughout the episode from
384 each other. We found that removing these non-discriminative signals (to the extent they behave as
385 additive offsets) via mean-centering helped reduce the embeddings’ anisotropy (i.e., increased the
386 range of measured between-vector similarities) and improved our ability to segment the episode
387 trajectory into meaningful events. Similarly, mean-centering both the episode and recalls before
388 relating their embedding representations helped mitigate broad-scale representational differences
389 driven by their diverging registers, and enabled us to better relate them based on event-specific,
390 content-driven patterns.

391 Our approach to mean-centering the recalls was guided by the particular latent representation

we wanted to remove versus retain. In segmenting each participant’s recall trajectory from each session into events (see *Event-segmentation*), our goal was to discriminate between embedding vectors strictly within a single recall sequence in order to identify temporally transient content-related patterns. We therefore centered each recall trajectory (in addition to the episode) on its own centroid to remove non-discriminative representations (e.g., register, broad-scale thematic content) and maximize the contrast between vectors within that single sequence. However, the analyses shown in Figures **REF TO DTW FIG** and **REF TO RDP FIG** entailed comparing sequences of recall events not only to the episode, but also to each other, both within- and across-participants, to examine whether within-participant similarities tended to exceed across-participant similarities. Here our goal was to remove representational artifacts that could work to bias our results towards significance. Specifically, participants’ idiolects (i.e., particular speaking styles and sets of words with which they describe the episode) will influence the embedding representations of their recalls in ways that tend to be similar between their immediate and delayed recalls, but different across participants. To correct for this confound, we jointly centered each participant’s set of immediate and delayed recall events on the two sequences’ shared centroid (i.e., the average vector across all events from both sessions). This served to reduce the recall embeddings’ anisotropy (by removing signals driven by high-level themes of the episode they recalled) and mitigate the effects of the spoken-recall register and participant-specific idiolect (both of which are shared between a participant’s recalls from the two sessions) while preserving any genuine drift in the high-level content of each participant’s recall over the delay (which per-session mean-centering would instead discard).

Event-segmentation

After constructing trajectories for the episode and each participant’s immediate and delayed recall, we used hidden Markov models (HMMs; Baum and Petrie, 1966; Rabiner, 1989) to segment each trajectory into a sequence of discrete events. Given a number of states K , an HMM models an observed time series as a progression through K hidden states, each of which has a characteristic signature in the data (i.e., a pattern that timepoints belonging to that state tend to exhibit) and

tends to persist across timepoints with occasional transitions rather than fluctuating freely. Because states persist, inferring the most probable state at each timepoint segments the observed time series into a sequence of events—i.e., spans of contiguous timepoints with the same most-probable state.

We segmented the episode trajectory into events using the HMM implemented in the BrainIAK toolbox (Kumar et al., 2021), which jointly learns a mean pattern for each of K states and the set of transitions between them via the Baum-Welch algorithm (Baum et al., 1970), with the added constraint that each state is visited exactly once in a strict left-to-right order (Baldassano et al., 2017). To select an appropriate number of states (i.e., events) for the episode, we used an optimization procedure we developed in prior work (Heusser et al., 2021) that aims to identify the single timescale of event representations emphasized most heavily in the episode’s narrative structure. We began by segmenting the (mean-centered) episode trajectory into K events for each K between 2 and 50, inclusive. For each candidate segmentation, we then computed the cosine similarity between each pair of timepoints assigned to the same event, and between each pair of timepoints assigned to different events. When constructing these distributions of within- and across-event similarities, we restricted comparisons to pairs of timepoints separated by no more than the duration of the longest event in the candidate segmentation, so that the within- and across-event distributions were computed over comparable temporal distances and the across-event distribution was not dominated by distant timepoint pairs that we would expect to be dissimilar regardless of event structure. We then selected the candidate segmentation that maximized the Wasserstein distance between these within- and across-event similarity distributions. We found that the maximally distinctive and internally self-similar segmentation was yielded by $K = 20$ events. **(TODO: ADD SUPP FIG SHOWING WASSERSTEIN DIST AS FUNCTION OF K)** Finally, we averaged the embedding vectors over timepoints between each pair of adjacent event boundaries to obtain a sequence of stable event-average representations for the episode.

To segment participants’ recall trajectories into sequences of events, we developed a novel HMM-driven procedure that grounded the recalls directly against the content of the episode participants viewed and recounted. Essentially, rather than re-learning separate sets of state representations from each participant’s recalls (which tend to be shorter, noisier approximations

447 of the episode content), we leveraged the state representations already learned from the episode to
 448 identify transitions between the episode events a given participant was (most probably) describing
 449 at a given point in time. First, we mean-centered the sequence of event-average embedding
 450 vectors for the episode, and each participant’s sequence of per-window embedding vectors from
 451 each session, separately. For each given recall trajectory, we then computed the cosine similarity
 452 between the embedding vector for each recall window and the event-average embedding vector
 453 for each of the 20 episode events. This yielded a number-of-windows by number-of-episode-
 454 events (20) similarity matrix. We then treated the episode events as the hidden states of a hidden
 455 Markov model with fixed parameters, and treated each recall window’s profile of similarities
 456 to the episode events, z-scored across all window-event pairs, as its distribution of emission
 457 scores over states. We imposed an ergodic transition structure (i.e., we allowed transitions from
 458 any state to any other state, since participants may recall the episode events out of order, and
 459 allowed states to be re-entered multiple times, since participants may revisit earlier events in
 460 their recalls) governed by separate additive transition penalties for forward-one transitions and
 461 all other transitions. Specifically, we imposed no penalty for remaining in the current state, a
 462 small penalty (1.5 normalized emissions units) for transitioning from a given state to the next
 463 (i.e., transitioning from recalling episode event n to recalling episode event $n + 1$), and a larger
 464 penalty (3.0 normalized emissions units) for all other transitions. Finally, we decoded the most
 465 probable sequence of episode-event labels using the Viterbi algorithm (Viterbi, 1967), collapsed
 466 runs of consecutive identically labeled recall windows into events, and averaged the embedding
 467 vectors over windows within each event. This procedure yielded, for each participant’s recall from
 468 each session, a sequence of event-average embedding vectors and a set of labels denoting which
 469 episode event each recall event corresponded to.

470 **Comparing trajectory structures**

471 We compared the geometric structures of the episode and recall event-sequences using dynamic
 472 time warping (DTW; Sakoe and Chiba, 1978; Vintsyuk, 1968). The DTW algorithm finds the optimal
 473 order-preserving alignment between two sequences of vectors with potentially different lengths or

sampling rates by locally “stretching” and “compressing” them in time to maximize their overall similarity. This alignment yields a “warp path” that maps each vector in each sequence onto one or more consecutive vectors in the other sequence with the minimum total distance between matched vectors. The summed or average distance between matched vectors along the warp path (the “warp cost”) is commonly taken as a measure of the two sequences’ structural dissimilarity. We computed the warp cost as the average cosine distance between DTW-matched events so that the cost values would be comparable across comparisons involving event-sequences of different lengths (whereas summed distance necessarily grows with sequence length).

Data availability

All of the data analyzed in this manuscript may be found at <https://github.com/ContextLab/memory-dynamics>.

Code availability

All of the code for running our experiment and carrying out the analyses may be found at <https://github.com/ContextLab/memory-dynamics>.

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